

Quantum Computing

Lecture $|02\rangle$

The Quantum Bit

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Outline

- Probabilistic algorithms
- The Quantum Bit (qubit)
- Qubit operations



Probabilistic Algorithms

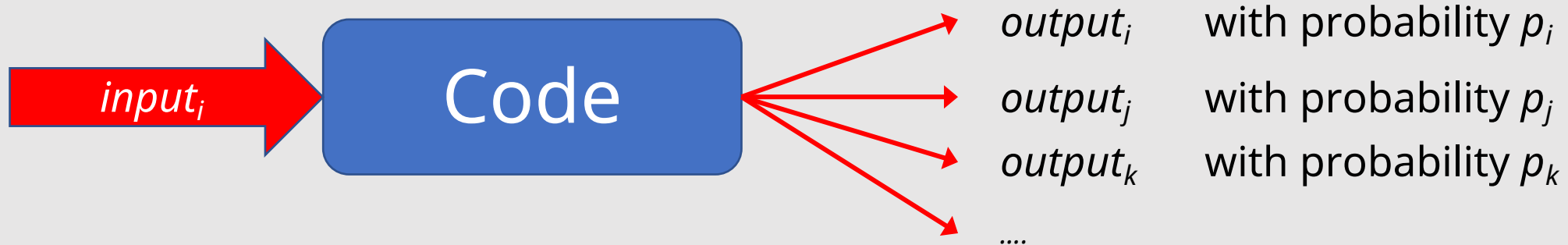
- Most of the algorithms you've seen so far are **deterministic**
- For **a given input** they'll produce the same output
 - (assuming your code doesn't crash or hangs 😊)



- **Note:** multiple input values a, b, c , etc. can “map” to the same output value
 - This is OK – think about a program for a simple calculator!

Probabilistic Algorithms

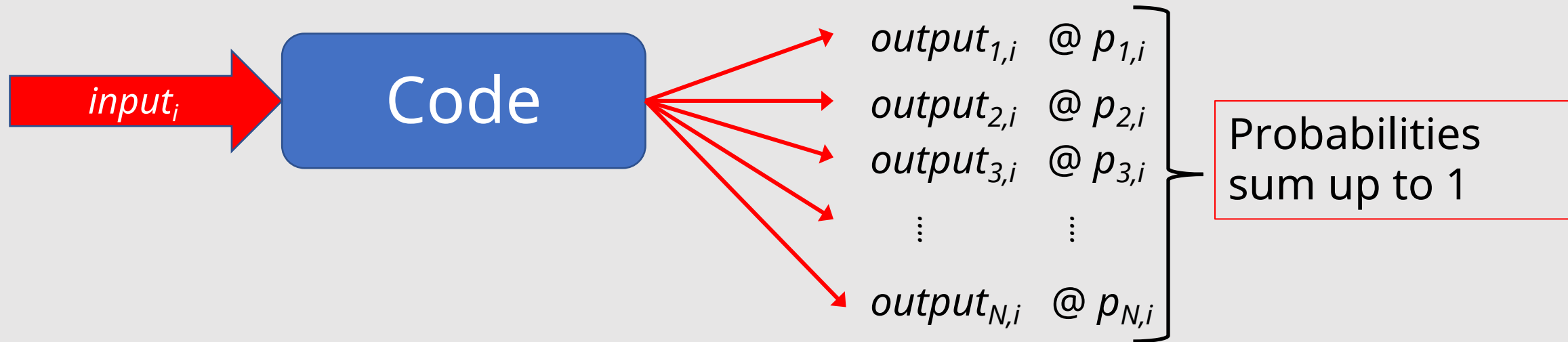
- With stochastic (probabilistic) algorithms it can be that



- We assume that the number of possible outputs is finite (say N)
 - but not necessarily so the number of inputs

Probabilistic Algorithms

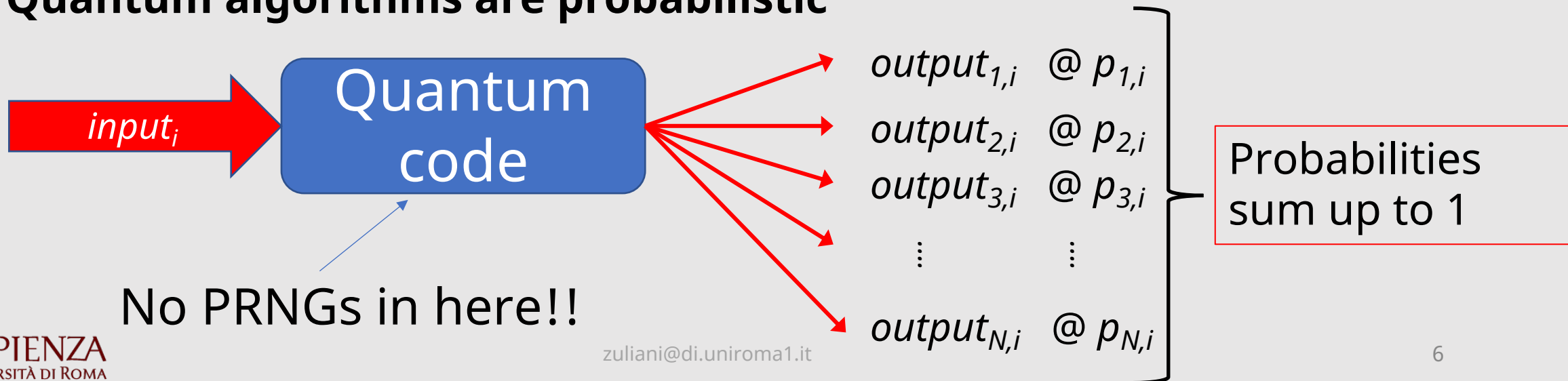
- For any input i we have at most N possible outputs, and they satisfy



- The program will output something with **probability 1**
 - Essentially, we are asking for (**probabilistic**) termination!

Probabilistic Algorithms

- Classical (non-quantum) algorithms introduce probabilistic behaviour via **pseudo-random number generators** (PRNGs)
 - PRNGs produce, say, a sequence of integers such that given an element of the sequence it is difficult to predict the next one (i.e., the best one can do is “to guess” randomly)
 - **Note:** PRNGs necessarily produce **finite** sequences: after many elements are produced, the sequence repeats itself (i.e., it is no longer random!)
- **Quantum algorithms are probabilistic**



The Quantum Bit (Qubit)

- A classical bit is a binary digit: it's either "0" or "1".
- The quantum equivalent are given by the **vectors**

$$\text{"0"} = |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{"1"} = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- There's more! The "true" state of a qubit is the **superposition**

$$\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \alpha |0\rangle + \beta |1\rangle$$

where α, β are **complex numbers** and satisfy $|\alpha|^2 + |\beta|^2 = 1$.

The Quantum Bit (Qubit)

$$\alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \alpha|0\rangle + \beta|1\rangle$$

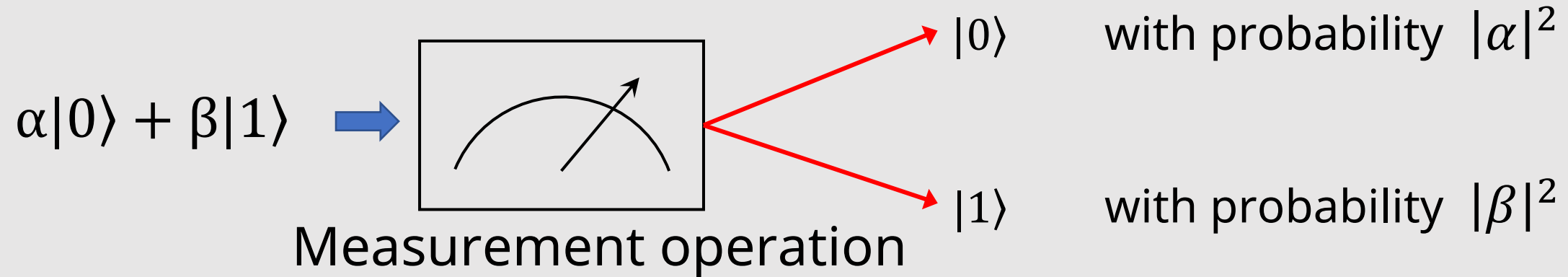
α, β are **probability amplitudes**. (Recall that $|\alpha|^2 + |\beta|^2 = 1$.)

Example:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

The Quantum Bit (Qubit)

- The “true” state of a qubit **CANNOT** be observed
 - We cannot in principle find out precisely the value of α, β
- Qubits are **measured**



- We know **for sure** that after measurement the qubit is (a multiple) either $|0\rangle$ or $|1\rangle$.

The Quantum Bit (Qubit)

- Note that for any complex α , $|i\alpha| = |\alpha|$.
- In general, $|z\alpha| = |\alpha|$ if $|z| = 1$. (Easy to prove.)
- But this means that for any complex $|z| = 1$, the qubit states

$$z\alpha|0\rangle + z\beta|1\rangle \quad (|\alpha|^2 + |\beta|^2 = 1)$$

CANNOT be distinguished by any measurement!!!

Qubit Operations

- Quantum transformations (*except* measurements) are **linear**

$$T(\alpha|0\rangle + \beta|1\rangle) = \alpha T(|0\rangle) + \beta T(|1\rangle)$$

- Any linear transformation (on a vector space) can be represented by a matrix
- The “do nothing” transformation is the 2x2 identity matrix

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$I|0\rangle = |0\rangle$$

$$I|1\rangle = |1\rangle$$

(Check that $I|\Phi\rangle = |\Phi\rangle$, where $|\Phi\rangle$ is the general qubit state above.)

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Qubit Operations: NOT

- The equivalent of the NOT gate on classical bits

$$\text{NOT} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \sigma_x$$

is one of the three Pauli matrices (much used in quantum physics and computation!)

$$\text{NOT} |0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

$$\text{NOT} |1\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

$$\Rightarrow \text{NOT NOT} = I$$

Qubit Operations: more Pauli Matrices

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma_y|0\rangle = i|1\rangle \quad \sigma_y|1\rangle = -i|0\rangle$$

$$\sigma_z|0\rangle = |0\rangle \quad \sigma_z|1\rangle = -|1\rangle$$

- Verify the equalities above!
- Exercise: What are σ_x^2 , σ_y^2 , σ_z^2 ?

Qubit Operations: Hadamard Transform

- Critically important in quantum computing:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

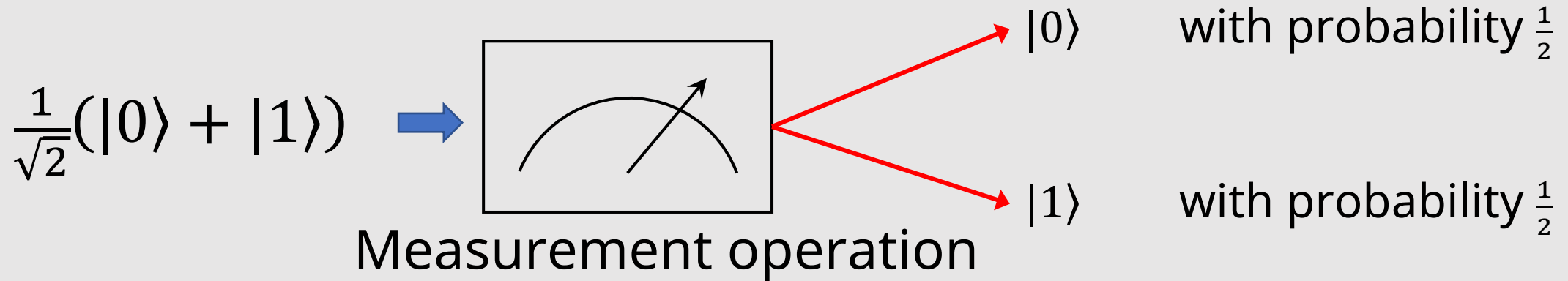
$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

- From a classical state we obtain a **superposition**!
- What happens if we measure it?

Qubit Operations: Hadamard Transform

- A “true”, random bits generator!

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$



(Exercise: try $H|1\rangle$.)

Quantum Evolution

- Is any matrix an allowed quantum transformation?
- NO! Matrices must preserve the **norm** of their input vectors

$$v = \alpha|0\rangle + \beta|1\rangle \quad \|v\| = \sqrt{|\alpha|^2 + |\beta|^2} \text{ is the norm of } v$$

- In quantum computing we already have $|\alpha|^2 + |\beta|^2 = 1$, so $\|v\| = 1$ for any qubit state v .
- Intuitively, after a measurement the qubit is in some state with probability 1.



Quantum Evolution

- These norm-preserving matrices are called **unitary**.

Definition: A matrix is called **unitary** if and only if

$$\|Uv\| = \|v\| \quad \text{for any qubit state } v$$

- Intuitively, unitary transforms “preserve the **probabilities**” (not necessarily the amplitudes!)
- (Check that the Pauli and Hadamard matrices are unitary.)

“Picturing” Qubits

- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with α, β complex and $|\alpha|^2 + |\beta|^2 = 1$
- [Polar coordinates: any complex number z can be written as $z = |z|(\cos \theta + i \sin \theta)$ for some angle θ .]
- By rewriting α and β in polar coordinates our qubit becomes

$$|\psi\rangle = e^{i\gamma}(\cos \frac{\theta}{2}|0\rangle + e^{i\varphi} \sin \frac{\theta}{2}|1\rangle)$$

where γ, φ and θ are *real* numbers.

- $|e^{i\gamma}| = 1$, so it has no observable effect and we may write

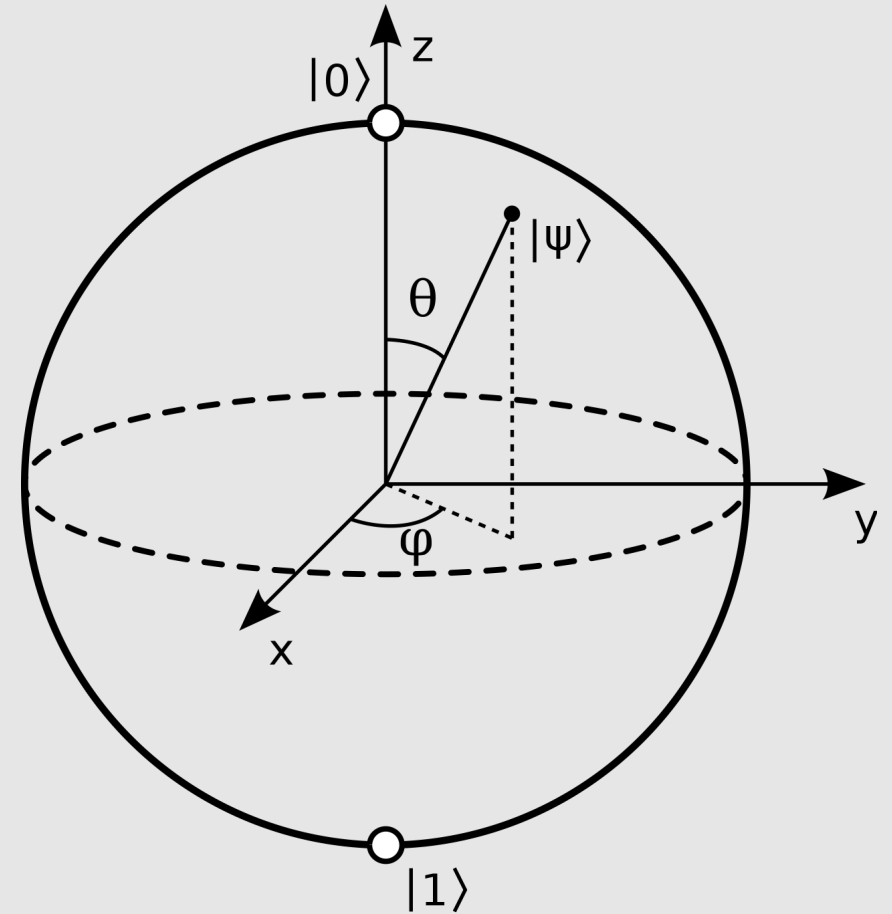
$$|\psi\rangle = (\cos \frac{\theta}{2}|0\rangle + e^{i\varphi} \sin \frac{\theta}{2}|1\rangle)$$

“Picturing” Qubits: the Bloch Sphere

$$|\psi\rangle = \left(\cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle\right)$$

where $0 < \theta < \pi$ and $0 < \varphi < 2\pi$

There is no simple Bloch sphere
equivalent for two or more qubits...



https://commons.wikimedia.org/wiki/File:Bloch_sphere.svg